

Laboratory work #2

Normalization

Task 1.

The Booking_receipt table violates 1NF because the seat_numbers attribute may contain several seat values in one cell, for example "12A, 12B". In 1NF every attribute must contain one atomic value. Because the table is not in 1NF, it also cannot yet be considered in 2NF or 3NF. After converting it to 1NF, additional partial and transitive dependencies must also be removed to reach 2NF and 3NF.

Task 2

Step 1 First normal form

Receipt_1NF(booking_id, passenger_full_name, passenger_passport_number, flight_number, departure_airport_name, departure_city, arrival_airport_name, arrival_city, airline_name, seat_number, ticket_price).

Primary key: (booking_id, seat_number). No foreign key here because the receipt is still one relation.

We change seat_numbers into seat_number. For example, booking 101 with seat_numbers = '12A, 12B' becomes one row for seat_number = '12A' and another for seat_number = '12B'. Both rows stay the same lead passenger, flight details and total ticket_price of 60000.00. Every seat value is now atomic. The table is in 1NF, but the booking details repeat. For example, the price is shown in both rows even though it is one total price for booking 101.

Step 2 Second normal form

To make the table 2NF, need to remove partial dependencies. The primary key in the 1NF table is (booking_id, passenger_passport_number). Some attributes depend only on booking_id, and passenger_full_name depends only on passenger_passport_number. Therefore, I divide the table into three tables. In 2NF, non-key attributes must depend on the whole primary key. Flight and booking information depend on booking_id, so I put them in the Booking table. Passenger_full_name depends on passenger_passport_number, so I put it in the Passenger table. Booking_Passenger connects the booking and passenger and stores the seat_number.

Booking(

booking_id ,

flight_number,

```

    departure_airport_name,
    departure_city,
    arrival_airport_name,
    arrival_city,
    airline_name,
    ticket_price
)
Passenger(
    passenger_passport_number ,
    passenger_full_name
)
Booking_Passenger(
    booking_id FK,
    passenger_passport_number FK,
    seat_number
)

```

Step 3 3NF

To make the tables 3NF, need to remove transitive dependencies. In 3NF, a non-key attribute should not depend on another non-key attribute.

booking_id-> flight_number

flight_number ->airline_name

flight_number -> departure_airport_name

departure_airport_name ->departure_city

flight_number->arrival_airport_name

arrival_airport_name->arrival_city

The flight information depends on flight_number, so I created the Flight table. The city depends on airport_name, so I created the Airport table. Now non-key attributes do not depend on other non-key attributes.

```

Booking(
booking_id ,
flight_number [FK],
ticket_price
)

Passenger(
passenger_passport_number ,
passenger_full_name
)

Booking_passenger(
booking_id FK,
passenger_passport_number [FK],
seat_number
)

Flight(
flight_number,
departure_airport_name [FK],
arrival_airport_name [FK],
airline_name
)

Airport(
Airport_name ,
City
)

```

The following functional dependencies were used:

booking_id → flight_number

booking_id → ticket_price

passenger_passport_number → passenger_full_name

flight_number → departure_airport_name

flight_number → arrival_airport_name

flight_number → airline_name

departure_airport_name → departure_city

arrival_airport_name → arrival_city

For 2NF, booking_id → flight_number, booking_id → ticket_price, and passenger_passport_number → passenger_full_name show partial dependencies on parts of the composite primary key.

For 3NF, dependencies such as booking_id → flight_number → airline_name and flight_number → departure_airport_name → departure_city show transitive dependencies. These dependencies were removed by creating separate Flight and Airport tables.

Task 4.

PRIMARY KEY : flight_id

Functional dependencies:

flight_id →

sch_departure_time,

sch_arrival_time,

departing_airport_id,

arriving_airport_id,

departing_gate,

arriving_gate,

airline_id,

act_departure_time,

act_arrival_time,

created_at,

updated_at

The departing_airport_id, arriving_airport_id and airline_id reference other tables, the airport and airline descriptive attributes are not stored in Flights. No non-key column in Flights is assumed to determine another non-key column in Flights. Therefore, the table is in 3NF.

Object Explorer PostgreSQL 18 postgres/postgres@PostgreSQL 18

Query

```
1 CREATE TABLE airline_info(  
2   airline_id INT PRIMARY KEY NOT NULL,  
3   airline_code VARCHAR(30) NOT NULL,  
4   airline_name VARCHAR(50) NOT NULL,  
5   airline_country VARCHAR(50) NOT NULL,  
6   created_at TIMESTAMP NOT NULL,  
7   updated_at TIMESTAMP NOT NULL,  
8   info VARCHAR(50) NOT NULL  
9 );  
10  
11 CREATE TABLE airport(  
12   airport_id INT PRIMARY KEY NOT NULL,  
13   airport_name VARCHAR(50) NOT NULL,  
14   country VARCHAR(50) NOT NULL,  
15   state VARCHAR(50) NOT NULL,  
16   city VARCHAR(50) NOT NULL,  
17   created_at TIMESTAMP NOT NULL,  
18   updated_at TIMESTAMP NOT NULL  
19 );  
20  
21 CREATE TABLE baggage_check(  
22   baggage_check_id INT PRIMARY KEY NOT NULL,  
23   check_result VARCHAR(50) NOT NULL,  
24   created_at TIMESTAMP NOT NULL,  
25   updated_at TIMESTAMP NOT NULL,  
26   booking_id INT NOT NULL,  
27   passenger_id INT NOT NULL  
28 );  
29  
30 CREATE TABLE baggage(  
31   baggage_id INT PRIMARY KEY NOT NULL,  
32   weight_in_kg DECIMAL(4,2) NOT NULL,  
33   created_at TIMESTAMP NOT NULL,  
34   updated_at TIMESTAMP NOT NULL,  
35   booking_id INT NOT NULL  
36 );  
37  
38 CREATE TABLE boarding_pass(  
39   boarding_pass_id INT PRIMARY KEY NOT NULL,  
40   booking_id INT NOT NULL,  
41   seat VARCHAR(50) NOT NULL,  
42   boarding_time TIMESTAMP NOT NULL,  
43   created_at TIMESTAMP NOT NULL,  
44   updated_at TIMESTAMP NOT NULL  
45 );
```

Total rows: Query complete 00:00:00.042 LF Ln 141, Col 36

Object Explorer PostgreSQL 18 postgres/postgres@PostgreSQL 18

Query

```
34 CREATE TABLE boarding_pass(  
35   boarding_pass_id INT PRIMARY KEY NOT NULL,  
36   booking_id INT NOT NULL,  
37   seat VARCHAR(50) NOT NULL,  
38   boarding_time TIMESTAMP NOT NULL,  
39   created_at TIMESTAMP NOT NULL,  
40   updated_at TIMESTAMP NOT NULL  
41 );  
42  
43 CREATE TABLE booking_flight(  
44   booking_flight_id INT PRIMARY KEY NOT NULL,  
45   booking_id INT NOT NULL,  
46   flight_id INT NOT NULL,  
47   created_at TIMESTAMP NOT NULL,  
48   updated_at TIMESTAMP NOT NULL  
49 );  
50  
51 CREATE TABLE booking(  
52   booking_id INT PRIMARY KEY NOT NULL,  
53   flight_id INT NOT NULL,  
54   passenger_id INT NOT NULL,  
55   booking_platform VARCHAR(50) NOT NULL,  
56   created_at TIMESTAMP NOT NULL,  
57   updated_at TIMESTAMP NOT NULL,  
58   status VARCHAR(50) NOT NULL,  
59   price DECIMAL(7,2) NOT NULL  
60 );  
61  
62 CREATE TABLE flights(  
63   flight_id INT PRIMARY KEY NOT NULL,  
64   sch_departure_time TIMESTAMP NOT NULL,  
65   sch_arrival_time TIMESTAMP NOT NULL,  
66   departing_airport_id INT NOT NULL,  
67   arriving_airport_id INT NOT NULL,  
68   departing_gate VARCHAR(50) NOT NULL,  
69   arriving_gate VARCHAR(50) NOT NULL,  
70   airline_id INT NOT NULL,  
71   act_departure_time TIMESTAMP NOT NULL,  
72   act_arrival_time TIMESTAMP NOT NULL,  
73   created_at TIMESTAMP NOT NULL,  
74   updated_at TIMESTAMP NOT NULL  
75 );
```

Total rows: Query complete 00:00:00.042 LF Ln 141, Col 36

Object Explorer PostgreSQL 18 postgres/postgres@PostgreSQL 18

Query

```
73 CREATE TABLE passengers(  
74     passenger_id INT PRIMARY KEY NOT NULL,  
75     first_name VARCHAR(50) NOT NULL,  
76     last_name VARCHAR(50) NOT NULL,  
77     date_of_birth DATE NOT NULL,  
78     gender VARCHAR(50) NOT NULL,  
79     country_of_citizenship VARCHAR(50) NOT NULL,  
80     country_of_residence VARCHAR(50) NOT NULL,  
81     passport_number VARCHAR(20) NOT NULL,  
82     created_at TIMESTAMP NOT NULL,  
83     updated_at TIMESTAMP NOT NULL  
84 );  
85 CREATE TABLE security_check(  
86     security_check_id INT PRIMARY KEY NOT NULL,  
87     check_result VARCHAR(20) NOT NULL,  
88     created_at TIMESTAMP NOT NULL,  
89     updated_at TIMESTAMP NOT NULL,  
90     passenger_id INT NOT NULL  
91 );  
92 ALTER TABLE airline_info RENAME TO airline;  
93 ALTER TABLE booking RENAME COLUMN price TO ticket_price;  
94 ALTER TABLE flights ALTER COLUMN departing_gate TYPE TEXT;  
95 ALTER TABLE airline DROP COLUMN info;  
96  
97 ALTER TABLE security_check ADD CONSTRAINT fk_security_check_passenger  
98 FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id);  
99  
100 ALTER TABLE booking ADD CONSTRAINT fk_booking_passenger  
101 FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id);  
102  
103 ALTER TABLE baggage_check ADD CONSTRAINT fk_baggage_check_passenger  
104 FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id);  
105  
106 ALTER TABLE baggage_check ADD CONSTRAINT fk_baggage_check_booking  
107 FOREIGN KEY (booking_id) REFERENCES booking(booking_id);  
108  
109 ALTER TABLE baggage ADD CONSTRAINT fk_baggage_booking  
110 FOREIGN KEY (booking_id) REFERENCES booking(booking_id);  
111
```

Total rows: Query complete 00:00:00.042 LF Ln 141, Col 36

Object Explorer PostgreSQL 18 postgres/postgres@PostgreSQL 18

Query

```
112 ALTER TABLE boarding_pass ADD CONSTRAINT fk_boarding_pass_booking  
113 FOREIGN KEY (booking_id) REFERENCES booking(booking_id);  
114  
115 ALTER TABLE booking_flight ADD CONSTRAINT fk_booking_flight_booking  
116 FOREIGN KEY (booking_id) REFERENCES booking(booking_id);  
117  
118 ALTER TABLE booking_flight ADD CONSTRAINT fk_booking_flight_flight  
119 FOREIGN KEY (flight_id) REFERENCES flights(flight_id);  
120  
121 ALTER TABLE flights ADD CONSTRAINT fk_flights_departing_airport  
122 FOREIGN KEY (departing_airport_id) REFERENCES airport(airport_id);  
123  
124 ALTER TABLE flights ADD CONSTRAINT fk_flights_arriving_airport  
125 FOREIGN KEY (arriving_airport_id) REFERENCES airport(airport_id);  
126  
127 ALTER TABLE flights ADD CONSTRAINT fk_flights_airline  
128 FOREIGN KEY (airline_id) REFERENCES airline(airline_id);  
129  
130 -- Normalization task 5  
131 CREATE TABLE receipt_passenger (  
132     passenger_passport_number VARCHAR(20) PRIMARY KEY NOT NULL,  
133     passenger_full_name VARCHAR(100) NOT NULL  
134 );  
135 CREATE TABLE receipt_airport (  
136     airport_name VARCHAR(50) PRIMARY KEY NOT NULL,  
137     city VARCHAR(50) NOT NULL  
138 );  
139 CREATE TABLE receipt_flight (  
140     flight_number VARCHAR(20) PRIMARY KEY,  
141     departure_airport_name VARCHAR(50),  
142     arrival_airport_name VARCHAR(50),  
143     airline_name VARCHAR(50),  
144     FOREIGN KEY (departure_airport_name) REFERENCES receipt_airport(airport_name),  
145     FOREIGN KEY (arrival_airport_name) REFERENCES receipt_airport(airport_name)  
146 );  
147 CREATE TABLE receipt_booking(  
148     booking_id INT PRIMARY KEY,  
149     flight_number VARCHAR(20),  
150     ticket_price DECIMAL(10,2) NOT NULL  
151 );
```

Total rows: Query complete 00:00:00.042 LF Ln 141, Col 36

Object Explorer PostgreSQL 18 postgres/postgres@PostgreSQL 18

Query

```
147 CREATE TABLE receipt_booking(  
148 booking_id INT PRIMARY KEY,  
149 flight_number VARCHAR(20),  
150 ticket_price NUMERIC(7,2),  
151 FOREIGN KEY(flight_number) REFERENCES receipt_flight(flight_number)  
152 );  
153 CREATE TABLE receipt_booking_seat (  
154 booking_id INT,  
155 passenger_passport_number VARCHAR(20),  
156 seat_number VARCHAR(10),  
157 PRIMARY KEY (booking_id,passenger_passport_number),  
158 FOREIGN KEY (booking_id) REFERENCES receipt_booking(booking_id),  
159 FOREIGN KEY (passenger_passport_number) REFERENCES receipt_passenger(passenger_passport_number)  
160 );  
161 INSERT INTO receipt_airport(airport_name,city)  
162 VALUES  
163 ('Almaty International Airport', 'Almaty'),  
164 ('Astana International Airport', 'Astana'),  
165 ('Shymkent International Airport', 'Shymkent'),  
166 ('Aktau International Airport', 'Aktau'),  
167 ('Atyrau International Airport', 'Atyrau');  
168 INSERT INTO receipt_flight  
169 (flight_number, departure_airport_name, arrival_airport_name, airline_name)  
170 VALUES  
171 ('KZ101', 'Almaty International Airport',  
172 'Astana International Airport', 'Air Astana'),  
173  
174 ('DV202', 'Astana International Airport',  
175 'Shymkent International Airport', 'SCAT Airlines'),  
176  
177 ('KZ303', 'Shymkent International Airport',  
178 'Aktau International Airport', 'Air Astana'),  
179  
180 ('DV404', 'Aktau International Airport',  
181 'Atyrau International Airport', 'SCAT Airlines'),  
182  
183 ('KZ789', 'Atyrau International Airport',  
184 'Almaty International Airport', 'Fly arystan');  
185 INSERT INTO receipt_booking(booking_id, flight_number, ticket_price)
```

Total rows: Query complete 00:00:00.042 LF Ln 141, Col 36

Object Explorer PostgreSQL 18 postgres/postgres@PostgreSQL 18

Query

```
147 CREATE TABLE receipt_booking(  
148 booking_id INT PRIMARY KEY,  
149 flight_number VARCHAR(20),  
150 ticket_price NUMERIC(7,2),  
151 FOREIGN KEY(flight_number) REFERENCES receipt_flight(flight_number)  
152 );  
153 CREATE TABLE receipt_booking_seat (  
154 booking_id INT,  
155 passenger_passport_number VARCHAR(20),  
156 seat_number VARCHAR(10),  
157 PRIMARY KEY (booking_id,passenger_passport_number),  
158 FOREIGN KEY (booking_id) REFERENCES receipt_booking(booking_id),  
159 FOREIGN KEY (passenger_passport_number) REFERENCES receipt_passenger(passenger_passport_number)  
160 );  
161 INSERT INTO receipt_airport(airport_name,city)  
162 VALUES  
163 ('Almaty International Airport', 'Almaty'),  
164 ('Astana International Airport', 'Astana'),  
165 ('Shymkent International Airport', 'Shymkent'),  
166 ('Aktau International Airport', 'Aktau'),  
167 ('Atyrau International Airport', 'Atyrau');  
168 INSERT INTO receipt_flight  
169 (flight_number, departure_airport_name, arrival_airport_name, airline_name)  
170 VALUES  
171 ('KZ101', 'Almaty International Airport',  
172 'Astana International Airport', 'Air Astana'),  
173  
174 ('DV202', 'Astana International Airport',  
175 'Shymkent International Airport', 'SCAT Airlines'),  
176  
177 ('KZ303', 'Shymkent International Airport',  
178 'Aktau International Airport', 'Air Astana'),  
179  
180 ('DV404', 'Aktau International Airport',  
181 'Atyrau International Airport', 'SCAT Airlines'),  
182  
183 ('KZ789', 'Atyrau International Airport',  
184 'Almaty International Airport', 'Fly arystan');  
185 INSERT INTO receipt_booking(booking_id, flight_number, ticket_price)
```

Total rows: Query complete 00:00:00.042 LF Ln 141, Col 36

Object Explorer PostgreSQL 18 postgres/postgres@PostgreSQL 18

Query

```
169 (flight_number, departure_airport_name, arrival_airport_name, airline_name)
170 VALUES
171 ('KZ101', 'Almaty International Airport',
172      'Astana International Airport', 'Air Astana'),
173
174 ('DV202', 'Astana International Airport',
175      'Shymkent International Airport', 'SCAT Airlines'),
176
177 ('KZ303', 'Shymkent International Airport',
178      'Aktau International Airport', 'Air Astana'),
179
180 ('DV404', 'Aktau International Airport',
181      'Atyrau International Airport', 'SCAT Airlines'),
182
183 ('KZ789', 'Atyrau International Airport',
184      'Almaty International Airport', 'Fly arystan');
185 INSERT INTO receipt_booking(booking_id, flight_number, ticket_price)
186 VALUES
187 (1001, 'KZ101', 96000.00),
188 (1002, 'DV202', 52000.00),
189 (1003, 'KZ303', 61000.00),
190 (1004, 'DV404', 47000.00),
191 (1005, 'KZ789', 58000.00);
192 INSERT INTO receipt_passenger(passenger_passport_number, passenger_full_name)
193 VALUES
194 ('P100', 'Galym Guliya'),
195 ('P200', 'Maria Sailaubai'),
196 ('P300', 'Alina Hantarbai'),
197 ('P400', 'Galym Kyzzyrhan'),
198 ('P500', 'Dana Bekenova');
199 INSERT INTO receipt_booking_seat(booking_id, passenger_passport_number, seat_number)
200 VALUES
201 (1001, 'P100', '12A'),
202 (1001, 'P200', '12B'),
203 (1002, 'P300', '7C'),
204 (1003, 'P400', '18A'),
205 (1004, 'P500', '9F'),
206 (1005, 'P100', '4D');
207
```

Query History

Total rows: Query complete 00:00:00.042 LF Ln 141, Col 36